

Polar Factorization and Monotone Rearrangement of Vector-Valued Functions

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ABSTRACT

We establish a fundamental polar factorization theorem for vector-valued maps, generalizing the classical polar decomposition of matrices to measure-preserving transformations. Given a probability measure and a cost function $c(x, y) = 1/2 |x - y|^2$, the optimal transport map T pushing μ to ν is the gradient of a convex potential: $T = \text{grad } \Phi$. In 1D, the optimal transport map is uniquely characterized as the monotone non-decreasing rearrangement. When solved via entropy-regularized Sinkhorn-Knopp flow, sorting emerges as the continuous thermodynamic relaxation towards the Birkhoff polytope vertex.

1. Monge-Kantorovich Formulation

In 1781, Gaspard Monge formulated the problem of transporting a mass distribution μ to ν with minimum mechanical work: $\min_T \int c(x, T(x)) d\mu(x)$.

Kantorovich (1942) relaxed this to couplings γ in $\Pi(\mu, \nu)$: $\min_\gamma \int c(x, y) d\gamma(x, y)$.

2. Brenier's Theorem

Theorem (Brenier 1991): If μ is absolutely continuous with compact support and $c(x, y) = 1/2 |x - y|^2$, there exists a unique optimal transport map T , and T is the gradient of a convex function Φ : $T = \text{grad } \Phi$.

In one dimension ($d = 1$), convex functions have non-decreasing derivatives (Φ' is monotone). Thus, Brenier's theorem guarantees that the unique optimal transport map in 1D is the monotone sorting function!

3. Entropic Regularization & Sinkhorn Flow

To compute optimal transport continuously, Cuturi (2013) added entropic regularization:

$$\min_P - \epsilon \log Z(P) \text{ subject to } P \mathbf{1} = \mathbf{1}/n, P^T \mathbf{1} = \mathbf{1}/n$$

The solution is $P^* = \text{diag}(u) K \text{diag}(v)$ where $K = \exp(-C / \epsilon)$. The iterative Sinkhorn-Knopp balancing updates converge exponentially fast to the optimal permutation matrix, realizing sorting as an entropic heat flow on the Birkhoff polytope.